

Variability of 7K and 11K SomaScan Plasma Proteomics Assays

Julián Candia,* Giovanna Fantoni, Francheska Delgado-Peraza, Nader Shehadeh, Toshiko Tanaka, Ruin Moaddel, Keenan A. Walker, and Luigi Ferrucci



Cite This: *J. Proteome Res.* 2024, 23, 5531–5539



Read Online

ACCESS |



Metrics & More



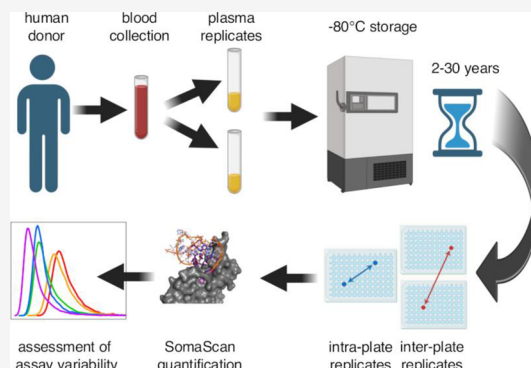
Article Recommendations



Supporting Information

ABSTRACT: SomaScan is an aptamer-based proteomics assay designed for the simultaneous measurement of thousands of human proteins with a broad range of endogenous concentrations. The 7K SomaScan assay has recently been expanded into the new 11K version. Following up on our previous assessment of the 7K assay, here, we expand our work on technical replicates from donors enrolled in the Baltimore Longitudinal Study of Aging. By generating SomaScan data from a second batch of technical replicates in the 7K version as well as additional intra- and interplate replicate measurements in the new 11K version using the same donor samples, this work provides useful precision benchmarks for the SomaScan user community. Beyond updating our previous technical assessment of the 7K assay with increased statistics, here, we estimate interbatch variability, assess inter- and intraplate variability in the new 11K assay, compare the observed variability between the 7K and 11K assays (leveraging the use of overlapping pairs of technical replicates), and explore the potential effects of sample storage time (ranging from 2 to 30 years) in the assays' precision.

KEYWORDS: SomaScan, SomaLogic, SOMAmers, 7K, 11K, plasma, proteomics, coefficient of variation, precision, variability, replicability, reproducibility



INTRODUCTION

SomaScan¹ is a highly multiplexed, aptamer-based assay capable of simultaneously measuring thousands of human proteins broadly ranging from femto- to micromolar concentrations. This technology relies on protein-capture SOMAmer (*Slow Offrate Modified Aptamer*) reagents,² designed to optimize high affinity, slow off-rate, and high specificity to target proteins. These targets extensively cover major molecular functions including receptors, kinases, growth factors, and hormones and span a diverse collection of secreted, intracellular, and extracellular proteins or domains. In recent years, SomaScan has increasingly been adopted as a powerful tool to discover biomarkers across a wide range of diseases and conditions as well as to elucidate their biological underpinnings in proteomics and multiomics studies.^{3–10}

Concurrently with its wider adoption, SomaScan has expanded its proteome coverage by increasing the number of SOMAmers included in different versions of the assay, from roughly 800 SOMAmers in 2009 to 1100 in 2012, 1300 in 2015, 5000 in 2018, 7000 in 2020, and the most recent 11,000 protein assay released in November 2023.¹¹ The first independent analysis of SomaScan normalization procedures and their technical variability was published by our teams at the U.S. National Institutes of Health (NIH) on the 1.1K and 1.3K assays,¹² later followed by the reports from other laboratories^{13–17} and by an updated assessment based on the 7K assay.¹⁸ In the latter, we utilized interplate technical

duplicates obtained from 102 human subjects enrolled in the Baltimore Longitudinal Study of Aging (BLSA), which allowed us to characterize different normalization procedures, evaluate assay variability, and provide detailed performance assessments.

In this work, we expand the scope of our previous study with the following aims: (i) to update the technical assessment of the 7K assay with the addition of a second batch of technical replicates, which allows us to increase statistics and to estimate interbatch variability; (ii) to extend the assessment to the recently released 11K assay using inter- and intraplate technical replicates; (iii) to compare the observed variability between the 7K and 11K assays, leveraging the use of overlapping pairs of technical replicates across the 7K and 11K assay versions; and (iv) to assess the potential relationship between technical variability and the time between sample collection and measurement over a time period spanning from 2 to 30 years.

Received: August 5, 2024

Revised: September 25, 2024

Accepted: October 21, 2024

Published: October 30, 2024



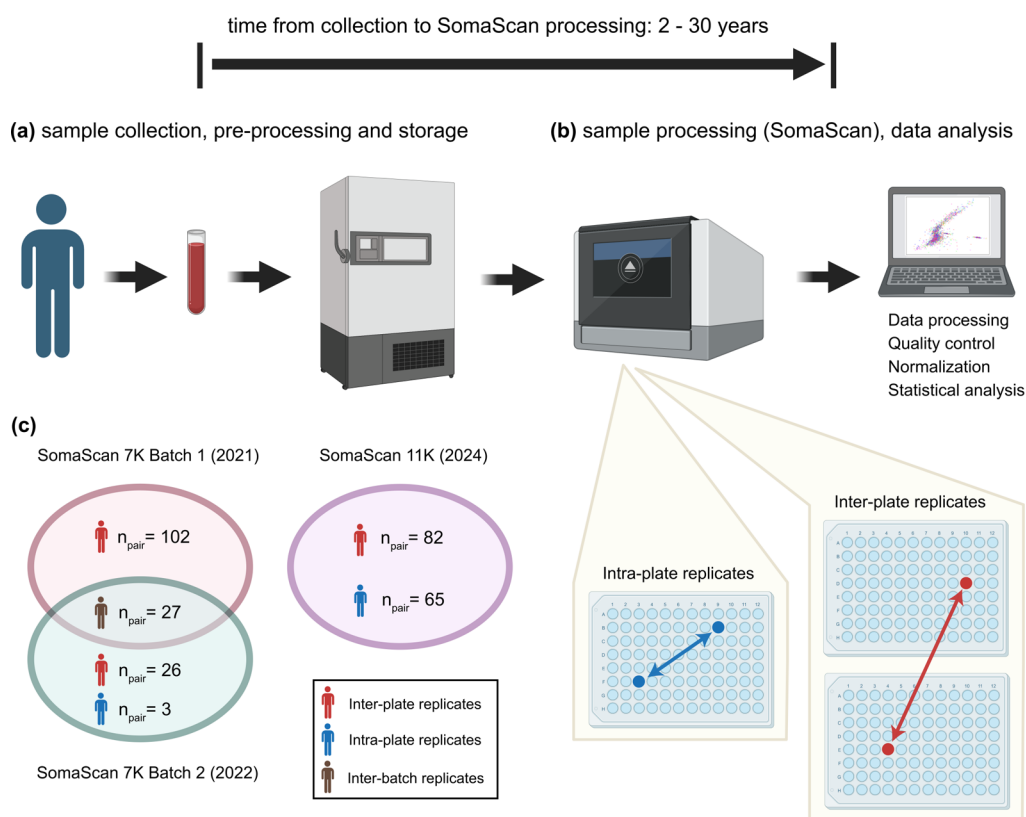


Figure 1. Schematic summary of the study. (a) Blood samples were collected from participants in the Baltimore Longitudinal Study of Aging (BLSA) and then preprocessed and stored at -80°C . (b) After a storage period of 2–30 years, samples were processed in SomaScan and analyzed. Some samples were run as intra- or interplate replicates. (c) Replicate pairs were run in two separate SomaScan 7K batches in years 2021 and 2022, respectively, followed by SomaScan 11K in year 2024. All of the replicate samples run in the 11K assay were previously run in the 7K assay.

EXPERIMENTAL SECTION

Study Design

Figure 1 shows the details of the study design. Blood samples were collected from participants in the Baltimore Longitudinal Study of Aging¹⁹ (BLSA), a study of normative human aging established in 1958 and conducted by the National Institute on Aging, NIH. The study protocol was conducted in accordance with Declaration of Helsinki principles and was reviewed and approved by the Institutional Review Board of the NIH's Intramural Research Program. Written informed consent was obtained from all of the participants. Following preprocessing using standard protocols, EDTA plasma vials were stored at -80°C (panel (a)). After a period of 2–30 years of storage, samples were processed in SomaScan and analyzed (panel (b)). Samples were organized in 96-well (8×12) plates. In our design, we included intra- and interplate technical replicate pairs, as indicated in panel (c). In order to allow for a paired analysis across assay versions, all of the replicate samples run on the 11K platform were previously run on the 7K platform.

SomaScan Assay

Plasma proteomic profiles were characterized using the 7K (v4.1) and 11K (v5.0) SomaScan assays (SomaLogic, Inc.; Boulder, CO, USA). These assays consist of 7,289 and 10,776 SOMAmers targeting annotated human proteins, respectively. All of the SOMAmers in the 7K assay were included in the new 11K version. In order to cover a broad range of endogenous concentrations, SOMAmers are binned into different dilution groups, namely, 20% (1:5) dilution for proteins typically observed in the femto- to pico-molar range (which comprise

about 80% of all human protein SOMAmers in the assays), 0.5% (1:200) dilution for proteins typically present in nanomolar concentrations (slightly below 20% of human protein SOMAmers in the assays), and 0.005% (1:20,000) dilution for proteins in micromolar concentrations (about 2–3% of human protein SOMAmers in the assays). The human plasma volume required is 55 μL per sample.

The experimental procedure follows a sequence of steps, namely: (1) SOMAmers are synthesized with a fluorophore, photocleavable linker, and biotin; (2) diluted samples are incubated with dilution-specific SOMAmers bound to streptavidin beads; (3) unbound proteins are washed away, and bound proteins are tagged with biotin; (4) UV light breaks the photocleavable linker, releasing complexes back into solution; (5) nonspecific complexes dissociate, while specific complexes remain bound; (6) a polyanionic competitor is added to prevent rebinding of nonspecific complexes; (7) biotinylated proteins (and bound SOMAmers) are captured on new streptavidin beads; and (8) after SOMAmers are released from the complexes by denaturing the proteins, fluorophores are measured following hybridization to complementary sequences on a microarray chip. The fluorescence intensity detected on the microarray, measured in RFU (*Relative Fluorescence Units*), is assumed to reflect the amount of available epitope in the original sample.

Raw data, as obtained after aggregation from slide-based hybridization microarrays, exhibit intraplate nuisance variance due to differences in loading volume, leaks, washing conditions, etc., which is then compounded with interplate biases. In order to account for intra- and interplate effects, we

apply a normalization procedure consisting of a sequence of steps, whose main elements are hybridization normalization, median signal normalization, plate-scale normalization, and interplate calibration (Table 1). Each of these steps generates

Table 1. Summary of Normalization Steps

step	name	abbreviation
0	raw	raw
1	hybridization control normalization	hyb
2	median signal normalization on calibrators	hyb.msnCal
3	plate-scale normalization	hyb.msnCal.ps
4	interplate calibration	hyb.msnCal.ps.cal
5	median signal normalization on all sample types	hyb.msnCal.ps.cal.msnAll

scale factors at different levels: plate-specific, SOMAmer dilution group, SOMAmer-specific, sample type, and combinations thereof. Besides removing technical variability, these scale factors can be used as quality control flags at the plate, sample, and SOMAmer levels.⁷ For further details, see refs 11 and 18.

SOMAmers are uniquely identified by their “SeqId”, but the relation between SOMAmers and annotated proteins is not one-to-one. In the 11K assay, the 10,776 SOMAmers that target human proteins are mapped to 9,609 unique UniProt IDs, yielding a total of 10,893 unique SOMAmer-UniProt ID pairs. [Supplementary Data 1](#) provides the list of SOMAmer-UniProt ID pairs, along with protein annotations extracted from the UniProt Knowledgebase (UniProtKB), including length (number of amino acids in the canonical sequence), mass (molecular weight), pH dependence, Redox potential,

temperature dependence, protein interactions, and subcellular location.

Statistical Methods

Coefficient of Variation Estimates from Replicate Pairs. A standard metric to characterize assay variability is the coefficient of variation (CV) defined as $CV = (\sigma/\mu) \times 100\%$, i.e., the ratio between the standard deviation and the mean over a sufficiently large number of technical replicates. However, for technical duplicates, this definition is unstable because it relies on the calculation of standard deviations using just two data points per distribution (where each distribution describes replicate measurements from one subject-SOMAmer pair). Instead, we previously proposed a grid-search procedure that estimates CV from n_{dupl} technical duplicates,¹⁸ as follows.

The theoretical relationship between the coefficient of variation and the expected fold change (FC) ratio between repeated measurements was derived by Reed et al.²⁰ Assuming that two replicate measurements, RFU_1 and RFU_2 , are log-normally distributed random variables with the same mean and variance, we define the fold change $FC \equiv RFU_1/RFU_2$ to be restricted to $FC \geq 1$, without loss of generality. Then, the theoretical probability that these replicate measurements differ by a fold change equal or larger than FC is given by

$$p_{\text{theor}}(FC; CV) = 2\Phi\left\{\frac{-\log_e(FC)}{\sqrt{2\log_e((CV/100)^2 + 1)}}\right\} \quad (1)$$

where Φ is the cumulative standard normal distribution function parametrized by CV.

For each SOMAmer, FC values obtained from replicate measurements define an empirical probability distribution,

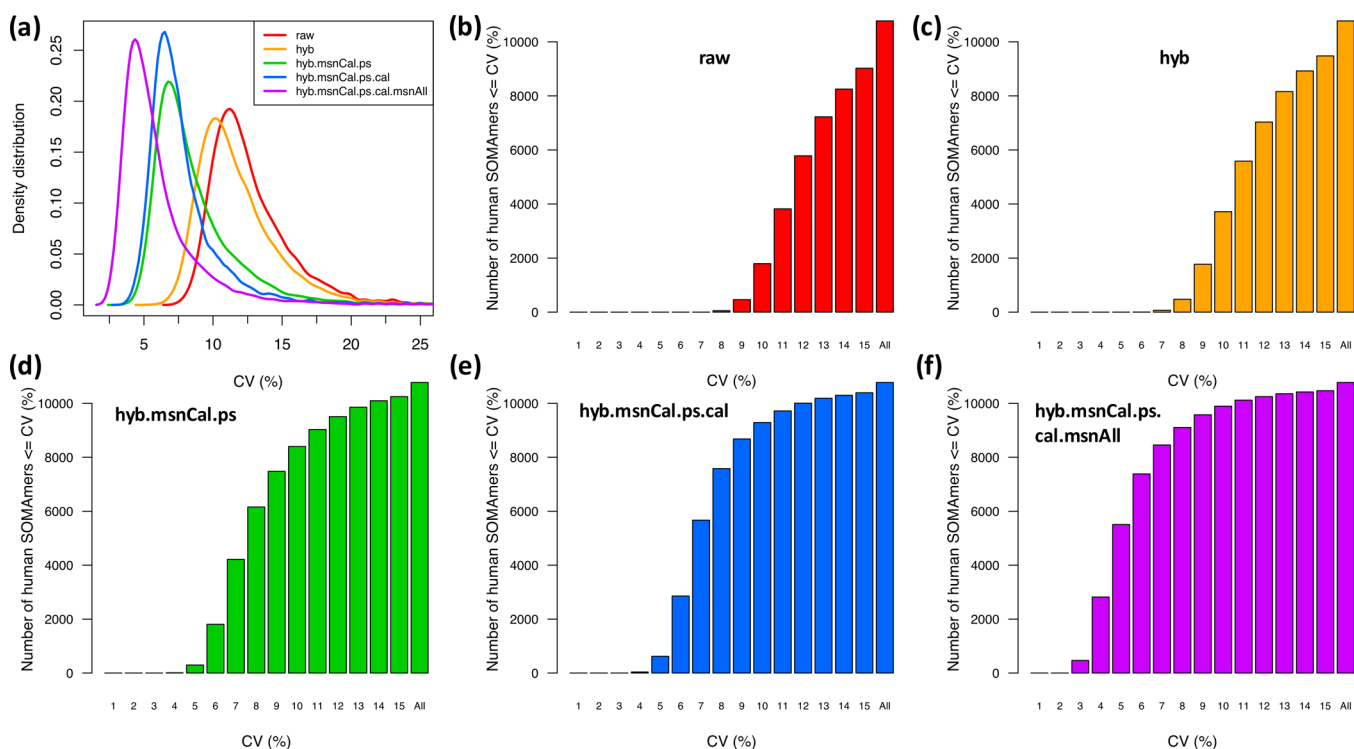


Figure 2. Distributions of coefficients of variation (CV). (a) Density distributions of CV across 10,776 human protein SOMAmers, obtained from interplate technical duplicates from 82 human subjects for different normalizations. (b)–(f) Cumulative distribution of the number of SOMAmers below a given CV for different normalizations, as indicated.

$p_{\text{emp}}(\text{FC})$, whose similarity with $p_{\text{theor}}(\text{FC};\text{CV})$ may be evaluated via the nonparametric, one-sample Kolmogorov–Smirnov (KS) test. Therefore, we may explore a dense ensemble of theoretical distributions generated by a pseudocontinuous scanning of the CV parameter in eq 1 and evaluate the similarity between the target empirical distribution and each theoretical one as the $-\log_{10}(p\text{-value})$ derived from the one-sample KS test between $p_{\text{emp}}(\text{FC})$ and $p_{\text{theor}}(\text{FC};\text{CV})$. Following this procedure, we thus estimate CV as the value that minimizes $-\log_{10}(p\text{-value})$ or, in other words, the CV value that generates the theoretical distribution most similar to the measured one.

Data and Software Availability

Data analysis was performed in R v.4.4.0. Protein annotations from UniProtKB were extracted using the R package queryup²¹ v.1.0.5. Plots were generated using R packages RColorBrewer v.1.1-3, viridis v.0.6.5, and calibrate v.1.7.7. The Abstract Graphic and Figure 1 were created with BioRender.com. Anonymized data sets and custom R source code used in our analyses are available on the Open Science Framework repository, <https://osf.io/2b9qe> (DOI 10.17605/OSF.IO/2B9QE).

RESULTS AND DISCUSSION

In order to explore the role of normalization in the 11K SomaScan assay, we used $n_{\text{dupl}} = 82$ interplate technical

Table 2. Summary of CV Percentile Distributions

CV percentile (%)	7K interplate	7K interbatch	11K interplate	11K intraplate
1	2.5	3	3	2
5	2.5	4	3.5	2
10	3	4.5	3.5	2.5
15	3	5	4	2.5
20	3.5	5.5	4	2.5
25	3.5	5.5	4	2.5
50	4.5	8	5	3
75	6	11.5	7	4
80	6.5	13	7.5	4
85	7	15	8.5	4.5
90	8.5	18.5	9.5	5
95	11	23.6	12	5.5
99	23	38	27	9.5

duplicates to implement our grid-search procedure (described in the Methods section) to all human protein SOMAmers. The resulting CV density distributions are shown in Figure 2a across all normalizations. Here, we observe that adding normalization steps shifts the distributions toward lower CV values. In greater detail, Figure 2b–f displays cumulative distributions of the number of SOMAmers below a given CV for each normalization, which are consistent with previous reports on interplate variability in the 7K SomaScan assay.¹⁸ The estimated CVs for each SOMAmer and normalization step are provided for the updated 7K interplate estimates (Supplementary Data 2), as well as the new estimates for 7K intrabatch (Supplementary Data 3), 11K interplate (Supplementary Data 4), and 11K intraplate (Supplementary Data 5). Unless stated otherwise, the remainder of this paper presents results for the fully normalized (“hyb.msnCal.ps.cal.msnAll”) data.

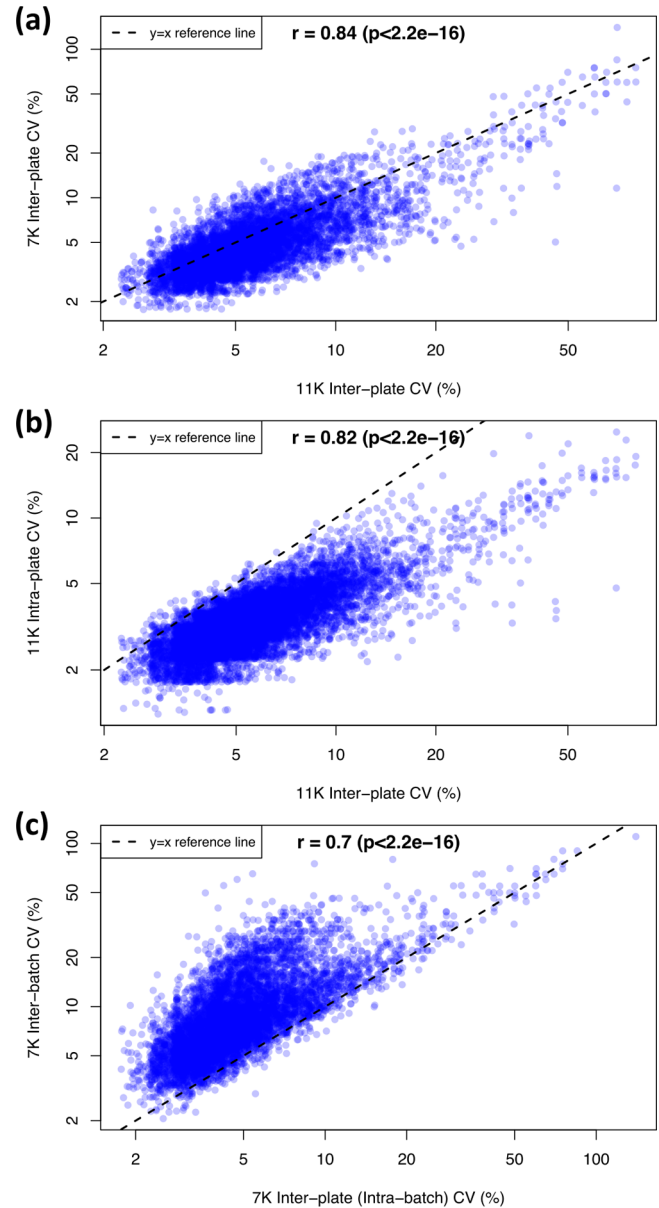


Figure 3. Comparison of coefficients of variation. (a) 7K interplate versus 11K interplate. (b) 11K intraplate versus 11K interplate. (c) 7K interbatch versus 7K interplate (intraplate). Pearson's correlation estimates and p-values are shown. Data have been fully normalized.

Table 2 shows a summary of the CV distributions by percentile for each case. On the one hand, we observe that interplate CV percentiles for 7K and 11K are similar (median CV = 4.5 and 5%, respectively). In greater detail, Figure 3a shows that CVs comprising 7289 human protein SOMAmers are strongly correlated. It is therefore reassuring that the overall precision of SomaScan has been maintained with the assay expansion from 7K to 11K. This is consistent with the findings of Rooney et al.²² when comparing between 5K and 11K SomaScan versions. On the other hand, we observe that intraplate variability is lower than interplate variability, as expected (median CV = 3 and 5%, respectively). As shown in Figure 3b, the correlation between intra- and interplate CV across 10,776 human protein SOMAmers remains strong, but the scatterplot cloud appears shifted in relation to the $y = x$ reference line (dashes). Finally, we observe that the precision

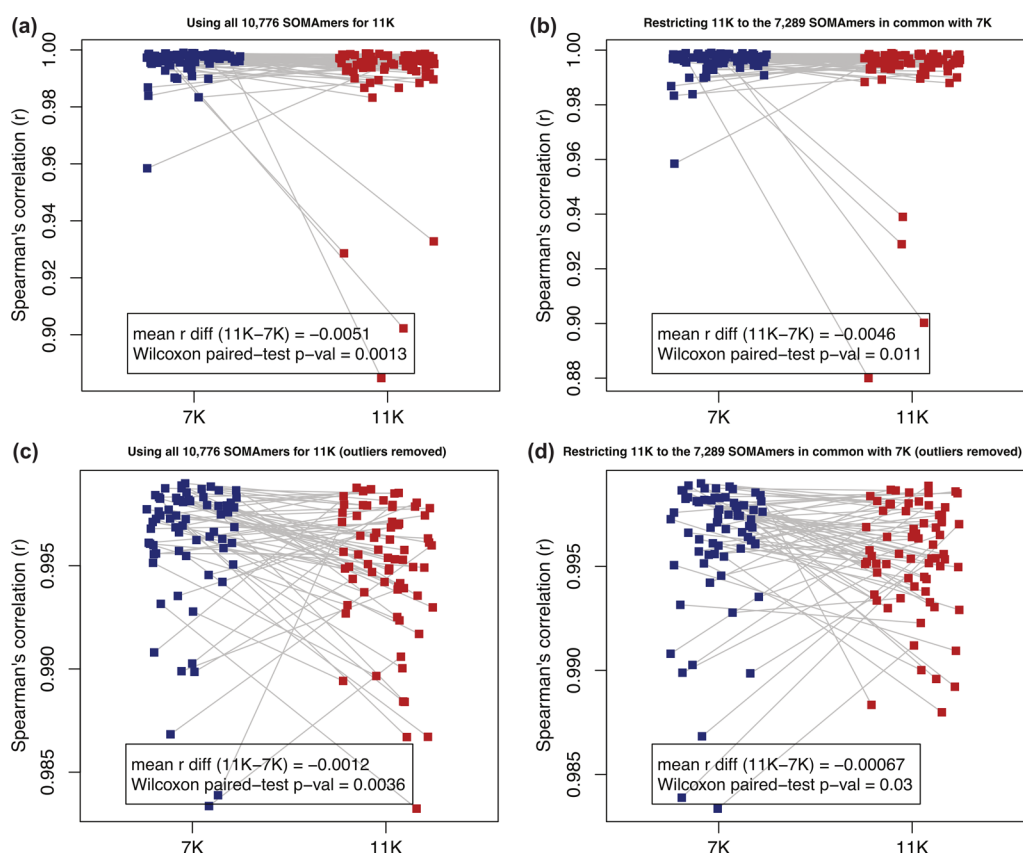


Figure 4. Interplate replicate correlations across SOMAmers: 11K vs 7K paired-sample comparison. For each interplate technical replicate, the Spearman's correlation across SOMAmers is computed using the 7K (blue) and 11K (red) assays. Matching samples across assays are joined by gray lines. Mean difference between correlation estimates in the 11K assay relative to the 7K is shown as well as the p -value from a Wilcoxon paired test. (a) Using 74 interplate technical replicates in each assay version (7289 human protein SOMAmers in the 7K assay and 10,776 human protein SOMAmers in the 11K assay). (b) Using 74 interplate technical replicates in each assay version (but restricting the 11K assay to the 7289 SOMAmers in common with 7K). (c,d) Same as panels (a,b), but with five outlier samples removed. Data have been fully normalized.

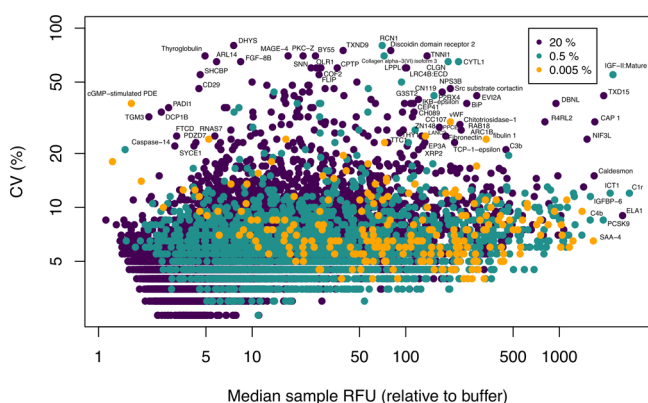


Figure 5. Brightness and variability of SOMAmers by dilution group. Interplate CV of 10,776 human protein SOMAmers as a function of the median signal-to-background ratio across experimental samples. SOMAmers are colored based on dilution: 20% ($n = 8644$), 0.5% ($n = 1911$), and 0.005% ($n = 221$). Data have been fully normalized.

across batches is significantly degraded (despite the fact that we renormalized both batches together using the same set of calibrator replicates). Indeed, the median (intrapatch) interplate CV = 4.5% is found to increase to median interbatch CV = 8. Correspondingly, Figure 3c shows that CVs are correlated but shifted toward larger values in the interbatch assessment relative to (intrapatch) interplate estimates.

One important design feature in our study is that all of the replicate samples run in the 11K assay were previously run in the 7K assay. More specifically, 74 interplate pairs were run in both assay versions, which allows us to perform a cross-version paired analysis of correlations across SOMAmers, as follows. For each replicate pair, we calculate the Spearman's correlation estimate across all SOMAmers. Figure 4a shows the correlations for all replicate pairs measured in the 7K (blue) and 11K (red) assay versions; matching samples across assays are joined by gray lines. The mean correlation estimate difference (defined as 11K minus 7K) is $\Delta r = -0.0051$, indicating that the correlation is slightly higher in the 7K assay relative to the 11K assay, with a Wilcoxon paired test p -value = 0.0013 . It should be noticed, however, that the 11K correlations in panel (a) were measured across all 10,776 human protein SOMAmers in the assay; in panel (b), we repeat the procedure but limiting 11K to the 7289 human protein SOMAmers in common with the 7K assay and observe again a small but significant negative difference between correlations in 11K and 7K. Since we notice five outlier samples, for which correlations in either assay appear unusually low, in panels (c,d), we repeated the analysis of panels (a,b) after removing these outliers. The conclusion, however, remains that the correlation across SOMAmers within interplate pairs appears to be slightly but significantly higher in the 7K SomaScan assay compared with the 11K assay version.

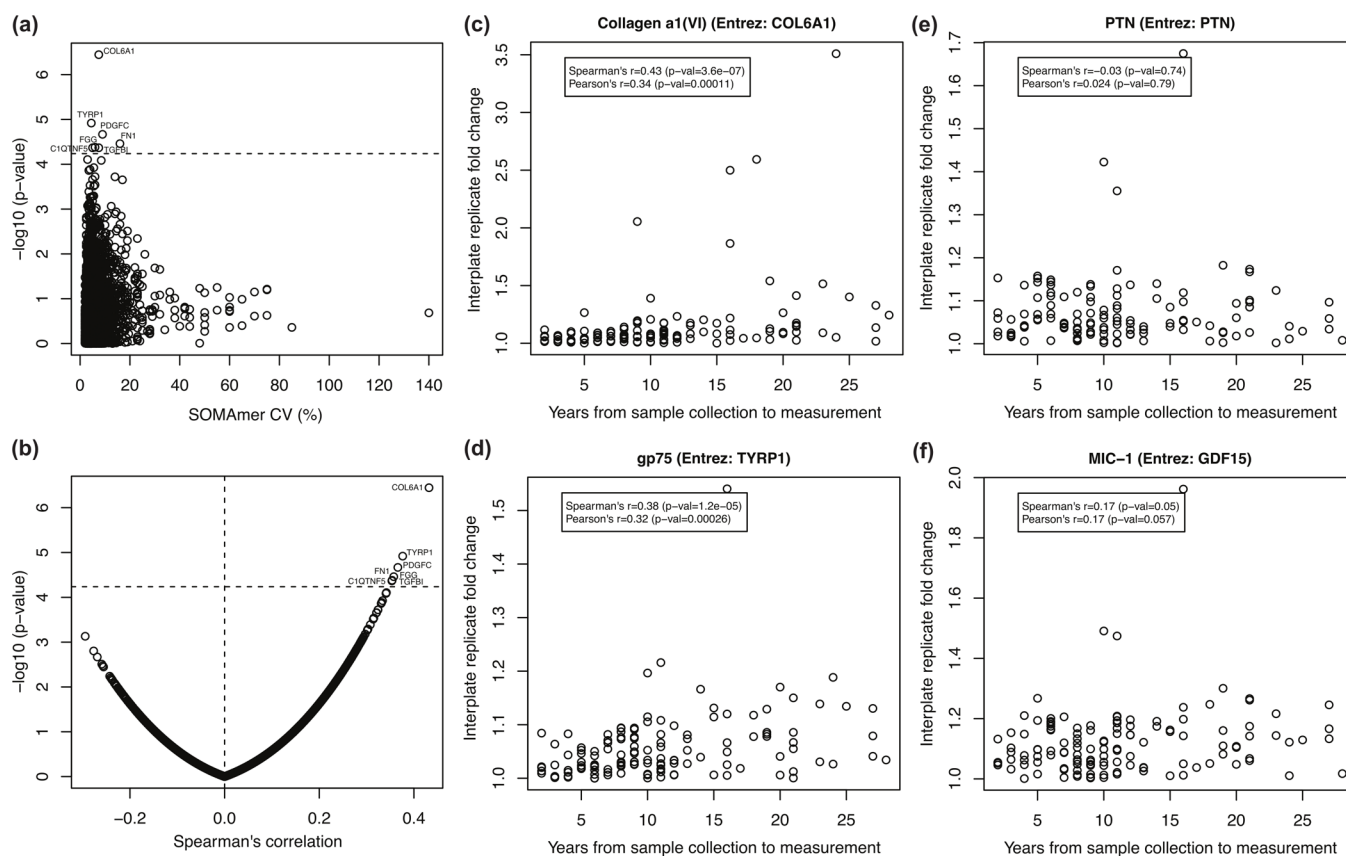


Figure 6. Association between technical variability and storage time in the 7K SomaScan assay. (a) Significance of the association between SOMAmer variability and storage time (y-axis) as a function of SOMAmer CV (x-axis). Dashed line shows the FDR-adjusted significance threshold $q = 0.05$. (b) Significance of the association between SOMAmer variability and storage time (y-axis) as a function of the Spearman's correlation (x-axis). (c–f) Interplate replicate fold change (FC) as a function of storage time. Spearman's and Pearson's correlation estimates and p-values are also shown. Panels (c,d) show the two SOMAMers with the strongest association (COL6A1 and TYRP1), whereas panels (e,f) showcase the two SOMAMers previously reported to exhibit the strongest association with aging (PTN and GDF15), which do not show significant effects with storage time. Data have been fully normalized.

Figure 5 shows the interplate CV as a function of the median signal-to-background ratio (SBR, defined as the median RFU of all samples divided by the median RFU of all buffer wells) for the 10,776 human protein SOMAMers in the 11K assay, colored by dilution group. The SBR estimates for each SOMAmer and normalization step are provided in [Supplementary Data 6](#). The Spearman's correlation between CV and SBR ($r = 0.48$) is primarily driven by the 20% dilution group ($r = 0.51$), which represents 80% of all human protein SOMAMers, followed by the 0.5% dilution group ($r = 0.13$), which represents 18% of all human protein SOMAMers. The third group, 0.005% dilution, comprises only 2% of all human protein SOMAMers and shows the opposite trend ($r = -0.17$). All correlation estimates are significant ($p < 0.05$). The CV median $\pm$ MAD across SOMAMers is $5 \pm 1.5\%$ (for all human protein SOMAMers as well as for the 20% dilution group), $6 \pm 2\%$ (for the 0.5% dilution group), and $7 \pm 2\%$ (for the 0.005% dilution group), respectively. The SBR median $\pm$ MAD across SOMAMers is 7.8 ± 7.1 (for all human protein SOMAMers), 6.4 ± 5.1 (for the 20% dilution group), 21.9 ± 22.3 (for the 0.5% dilution group), and 82.3 ± 103.0 (for the 0.005% dilution group), respectively. The pairwise Wilcoxon test comparisons between dilution groups indicate that all these CV and SBR differences are statistically significant (adjusted $p\text{-value} < 0.05$). We have not observed any significant associations between CV or SBR and the protein character-

istics described in [Supplementary Data 1](#) (such as protein length, mass, or number of protein interactions).

Finally, we explored the association between the technical variability and sample storage time. For the 7K SomaScan assay, 128 interplate pairs were measured between 2 and 28 years after the time of sample collection, preprocessing, and storage. For each SOMAmer, we calculated the correlation between the interplate replicate fold change and the sample storage time ([Supplementary Data 7](#)). This is summarized in [Figure 6a](#), which shows the p-value of the Spearman's correlation (transformed as $-\log_{10}(p\text{-value})$ on the y-axis) as a function of the SOMAmer CV (on the x-axis). The dashed line shows the FDR-adjusted significance threshold, $q = 0.05$. We observe that only seven proteins appear above the significance threshold. All of them are positively correlated, as shown by [Figure 6b](#), which is consistent with the interpretation that protein degradation is responsible for the increased variability over time. These effects are shown in greater detail in panels (c,d) for the top two proteins, namely, collagen type VI alpha 1 chain (COL6A1) and tyrosinase related protein 1 (TYRP1/gp75), respectively. Other proteins with significant degradation are related to plasma and coagulation. In contrast, two proteins reported to show the strongest association with human aging,⁵ namely, pleiotrophin (PTN) and macrophage inhibitory cytokine-1 (MIC-1/GDF15), show mild to insignificant degradation effects with

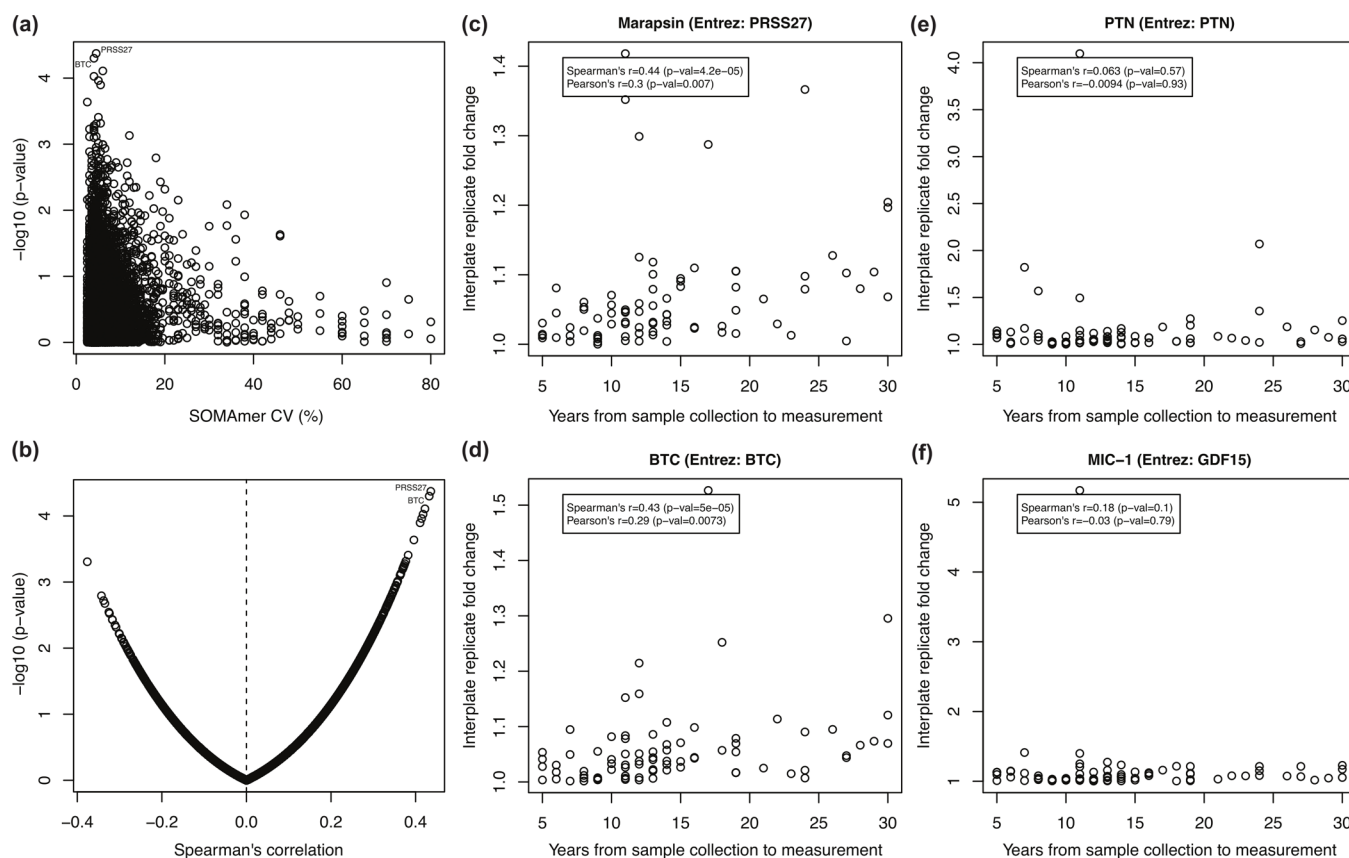


Figure 7. Association between technical variability and storage time in the 11K SomaScan assay. (a) Significance of the association between SOMAmer variability and storage time (y-axis) as a function of SOMAmer CV (x-axis). None of the observations reach the FDR-adjusted significance threshold $q = 0.05$. (b) Significance of the association between SOMAmer variability and storage time (y-axis) as a function of the Spearman's correlation (x-axis). (c–f) Interplate replicate fold change (FC) as a function of storage time. Spearman's and Pearson's correlation estimates and p-values are also shown. Panels (c,d) show the two SOMAmers with the strongest association (PRSS27 and BTC), whereas panels (e,f) showcase the two SOMAmers previously reported to exhibit the strongest association with aging (PTN and GDF15), which do not show significant effects with storage time. Data have been fully normalized.

storage time, as shown in panels (e,f), respectively. These observations have important consequences for the interpretation of findings from longitudinal studies such as BLSA, in which samples are typically stored for many years (and even decades) before they are thawed and processed to generate data.

Similarly, Figure 7 shows the association between technical variability and sample storage time for the 11K SomaScan assay, based on 82 interplate pairs stored for 5 to 30 years (Supplementary Data 8). None of the SOMAmers shows a significant degradation (using the $q = 0.05$ threshold after FDR adjustment), as shown in panel (a), although the top SOMAmers appear biased toward positive correlations (panel (b)), indicating that protein degradation may be occurring on these SOMAmer targets to a moderate degree. Data for the top two proteins, serine protease 27 (marapsin/PRSS27) and betacellulin (BTC), are shown in panels (c,d), respectively. Aging-associated proteins PTN and GDF15, shown in panels (e,f), display no significant degradation effects due to storage time. Many of the samples used in this study have not been aliquoted for single-use thawing, but we lack full historical records to track which of them may have been frozen and thawed multiple times. A separate study on the impact caused by repeated freeze–thaw cycles in 11K SomaScan quantification shows only 4% of affected SOMAmers,²³ the overlap between affected probes by storage (690 SOMAmers obtained

from Spearman's $p\text{-value} < 0.05$, Supplementary Data 8) and affected probes by freeze–thaw cycles (466 SOMAmers with $p\text{-value} < 0.05$ derived from mixed-effects models²³) is not statistically significant (31 overlapping SOMAmers, Fisher's exact test $p\text{-value} = 0.77$).

CONCLUSIONS

Similarly to previous versions of the SomaScan assay,^{12,18} the standard normalization procedure of 11K SomaScan data appears to operate as expected. While the median interplate CV of raw RFU counts was $\text{CV}_{\text{med}} = 12\%$, successive normalization steps reduced it after hybridization ($\text{CV}_{\text{med}} = 11\%$), plate-scale normalization ($\text{CV}_{\text{med}} = 8\%$), interplate calibration ($\text{CV}_{\text{med}} = 7\%$), and the fully normalized data including median signal normalization applied on all sample types ($\text{CV}_{\text{med}} = 5\%$). It is reassuring to observe no precision degradation in the assay expansion from the 7K version to the 11K version. Moreover, we observed a strong correlation of CV estimates across human protein SOMAmers between 7K and 11K.

Our findings have implications for the plate design of SomaScan studies. Intraplate variability ($\text{CV}_{\text{med}} = 3\%$) is significantly lower than interplate variability ($\text{CV}_{\text{med}} = 5\%$). Therefore, in order to prevent the occurrence of interplate effects in longitudinal studies, it is recommended to keep all longitudinal samples from the same subject together in the

same plate and randomize across plates based on relevant covariates (e.g., age and sex), which can be implemented with tools such as the R package OlinkAnalyze. Interbatch variability ($CV_{\text{med}} = 8\%$) is significantly higher than (intra-batch) interplate variability ($CV_{\text{med}} = 5\%$). The use of calibrators to account for interplate effects does not suffice to remove additional cross-batch effects. It is thus recommended that all samples in a study be run whenever possible within the same batch. Additional procedures for cross-batch correction may include the use of bridge samples to be added to runs across batches as well as the exploration of additional normalization procedures. Regarding the latter, bioinformatic tools developed to assess and remove batch effects for other omics may be useful in the context of SomaScan data processing, for instance: ComBat,²⁴ implemented in the sva R package,²⁵ a very well established method for batch correction in microarray and RNA-seq data; guided PCA;²⁶ and multi-MA normalization,²⁷ among others. This important topic certainly deserves further investigation.

Using fully normalized data, we find that correlations between interplate replicate pairs across all human protein SOMAmers are close to 1 for both 7K and 11K SomaScan assay versions. However, a paired comparison (allowed by the fact that we used overlapping pairs of technical replicates across both assay versions) reveals that the correlation is slightly but significantly higher in the 7K SomaScan assay than in the 11K assay version. Both assay versions exhibit a remarkable sensitivity, with 98% of the measured human proteins appearing >2-fold brighter in human plasma samples compared to buffer wells.

Finally, we explored the important question of the possible association between technical variability and sample storage time. If increased protein degradation was found to significantly affect the assay's precision, that would impose limits on the use of older samples stored in specimen banks, with particularly strong implications on the proteomic interrogation of long-established longitudinal studies such as the BLSA. We observed moderate degradation that appears to be protein-specific and not significantly associated with SOMAmers found to be affected by repeated freeze-thaw cycles;²³ hence, devising a correction process to account for this effect may be complex. Fortunately, we found that the variability increase in samples stored in a time frame from 2 to 30 years was not significant for most SOMAmers.

As SomaScan becomes more widely adopted and utilized as a state-of-the-art tool for proteomics discovery, we hope that this work will serve as a valuable technical reference and resource for the growing SomaScan user community.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00667>.

Protein annotations extracted from the UniProt Knowledgebase (UniProtKB) (XLSX)

7K SomaScan interplate coefficients of variation (XLSX)

7K SomaScan interbatch coefficients of variation (XLSX)

11K SomaScan interplate coefficients of variation (XLSX)

11K SomaScan intraplate coefficients of variation (XLSX)

11K SomaScan signal-to-background ratio estimates (XLSX)

7K SomaScan correlation between interplate replicate fold change and sample storage time (XLSX)

11K SomaScan correlation between interplate replicate fold change and sample storage time (XLSX)

Variability of 7K and 11K SomaScan plasma proteomics assays (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Julián Candia – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States; orcid.org/0000-0001-5793-8989; Email: julian.candia@nih.gov

Authors

Giovanna Fantoni – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Francheska Delgado-Peraza – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Nader Shehadeh – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Toshiko Tanaka – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Ruin Moaddel – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Keenan A. Walker – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Luigi Ferrucci – Intramural Research Program, National Institute on Aging, National Institutes of Health, Baltimore, Maryland 21224, United States

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00667>

Notes

The authors declare the following competing financial interest(s): Dr. Candia, Dr. Walker, and Dr. Ferrucci have given unpaid seminars and/or webinars sponsored or co-sponsored by SomaLogic.

■ ACKNOWLEDGMENTS

This research was supported entirely by the Intramural Research Program of the National Institute on Aging, NIH.

■ REFERENCES

- (1) Gold, L.; et al. Aptamer-based multiplexed proteomic technology for biomarker discovery. *PLoS ONE* **2010**, *5*, No. e15004.
- (2) Rohloff, J. C.; Gelinas, A. D.; Jarvis, T. C.; Ochsner, U. A.; Schneider, D. J.; Gold, L.; Janjic, N. Nucleic Acid Ligands With Protein-like Side Chains: Modified Aptamers and Their Use as Diagnostic and Therapeutic Agents. *Molecular Therapy Nucleic Acids* **2014**, *3*, No. e201.
- (3) Emilsson, V.; et al. Co-regulatory networks of human serum proteins link genetics to disease. *Science* **2018**, *361*, 769–773.

- (4) Sun, B. B.; et al. Genomic atlas of the human plasma proteome. *Nature* **2018**, *558*, 73.
- (5) Tanaka, T.; Biancotto, A.; Moaddel, R.; Moore, A. Z.; Gonzalez-Freire, M.; Aon, M. A.; Candia, J.; Zhang, P.; Cheung, F.; Fantoni, G.; CHI consortium; Semba, R. D.; Ferrucci, L. Plasma proteomic signature of age in healthy humans. *Aging Cell* **2018**, *17*, No. e12799.
- (6) Lehallier, B.; Gate, D.; Schaum, N.; Nanasi, T.; Lee, S. E.; Yousef, H.; Moran Losada, P.; Berdnik, D.; Keller, A.; Verghese, J.; Sathyan, S.; Franceschi, C.; Milman, S.; Barzilai, N.; Wyss-Coray, T. Undulating changes in human plasma proteome profiles across the lifespan. *Nature Medicine* **2019**, *25*, 1843–1850.
- (7) Williams, S. A.; et al. Plasma protein patterns as comprehensive indicators of health. *Nature Medicine* **2019**, *25*, 1851–1857.
- (8) Pietzner, M.; et al. Mapping the proteo-genomic convergence of human diseases. *Science* **2021**, *374*, No. eabj1541.
- (9) Roberts, J. A.; et al. A brain proteomic signature of incipient Alzheimer's disease in young APOE ϵ 4 carriers identifies novel drug targets. *Sci. Adv.* **2021**, *7*, No. eabi8178.
- (10) Walker, K. A.; et al. Large-scale plasma proteomic analysis identifies proteins and pathways associated with dementia risk. *Nature Aging* **2021**, *1*, 473–489.
- (11) Candia, J. *SomaScan Bioinformatics: Normalization, Quality Control, and Assessment of Pre-Analytical Variation*, 2024. <https://www.biorxiv.org/content/10.1101/2024.02.09.579724>.
- (12) Candia, J.; Cheung, F.; Kotliarov, Y.; Fantoni, G.; Sellers, B.; Griesman, T.; Huang, J.; Stuccio, S.; Zingone, A.; Ryan, B.; Tsang, J.; Biancotto, A. Assessment of Variability in the SOMAScan Assay. *Sci. Rep.* **2017**, *7*, 14248.
- (13) Kim, C.; Tworoger, S.; Stampfer, M.; Dillon, S.; Gu, X.; Sawyer, S.; Chan, A.; Libermann, T.; Eliassen, A. Stability and reproducibility of proteomic profiles measured with an aptamer-based platform. *Sci. Rep.* **2018**, *8*, 8382.
- (14) Tin, A.; et al. Reproducibility and Variability of Protein Analytes Measured Using a Multiplexed Modified Aptamer Assay. *Journal of Applied Laboratory Medicine* **2019**, *4*, 30–39.
- (15) Raffield, L.; Dang, H.; Pratte, K.; Jacobson, S.; Gillenwater, L.; Ampleford, E.; Barjaktarevic, I.; Basta, P.; Clish, C. Comparison of Proteomic Assessment Methods in Multiple Cohort Studies. *Proteomics* **2020**, *20*, No. 1900278.
- (16) Pietzner, M.; Wheeler, E.; Carrasco-Zanini, J.; Kerrison, N.; Oerton, E.; Koprulu, M.; Luan, J.; Hingorani, A.; Williams, S.; Wareham, N.; Langenberg, C. Synergistic insights into human health from aptamer- and antibody-based proteomic profiling. *Nat. Commun.* **2021**, *12*, 6822.
- (17) Dubin, R. F.; et al. Analytical and Biological Variability of a Commercial Modified Aptamer Assay in Plasma Samples of Patients with Chronic Kidney Disease. *J Appl Lab Med* **2023**, *8*, 491.
- (18) Candia, J.; Daya, G.; Tanaka, T.; Ferrucci, L.; Walker, K. Assessment of variability in the plasma 7k SomaScan proteomics assay. *Sci. Rep.* **2022**, *12*, 17147.
- (19) Ferrucci, L. The Baltimore Longitudinal Study of Aging (BLSA): a 50-year-long journey and plans for the future. *J Gerontol A Biol Sci Med Sci* **2008**, *63*, 1416–1419.
- (20) Reed, G.; Lynn, F.; Meade, B. Use of Coefficient of Variation in Assessing Variability of Quantitative Assays. *Clin. Diagn. Lab. Immunol.* **2002**, *9*, 1235–1239.
- (21) Voisinne, G. *queryup: Query the UniProt REST API using R*; Centre d'Immunologie de Marseille-Luminy, Aix Marseille Université, INSERM, CNRS: Marseille, France, 2019.
- (22) Rooney, M. R.; Chen, J.; Ballantyne, C. M.; Hoogeveen, R. C.; Boerwinkle, E.; Yu, B.; Walker, K. A.; Schlosser, P.; Selvin, E.; Chatterjee, N.; Couper, D.; Grams, M. E.; Coresh, J. *Plasma proteomic comparisons change as coverage expands for SomaLogic and Olink*, 2024. <https://www.medrxiv.org/content/10.1101/2024.07.11.24310161>.
- (23) Candia, J.; Fantoni, G.; Moaddel, R.; Delgado-Peraza, F.; Shehadeh, N.; Tanaka, T.; Ferrucci, L. *Effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using the SomaScan and Olink assays*, 2024. <https://www.biorxiv.org/content/10.1101/2024.09.21.613295>.
- (24) Johnson, W.; Li, C.; Rabinovic, A. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics* **2007**, *8*, 118.
- (25) Leek, J.; Johnson, W.; Parker, H.; Jaffe, A.; Storey, J. The sva package for removing batch effects and other unwanted variation in high-throughput experiments. *Bioinformatics* **2012**, *28*, 882.
- (26) Reese, S. E.; Archer, K. J.; Therneau, T. M.; Atkinson, E. J.; Vachon, C. M.; de Andrade, M.; Kocher, J.-P. A.; Eckel-Passow, J. E. A new statistic for identifying batch effects in high-throughput genomic data that uses guided principal component analysis. *Bioinformatics* **2013**, *29*, 2877.
- (27) Hong, M.-G.; Lee, W.; Nilsson, P.; Pawitan, Y.; Schwenk, J. Multidimensional Normalization to Minimize Plate Effects of Suspension Bead Array Data. *J. Proteome Res.* **2016**, *15*, 3473–3480.